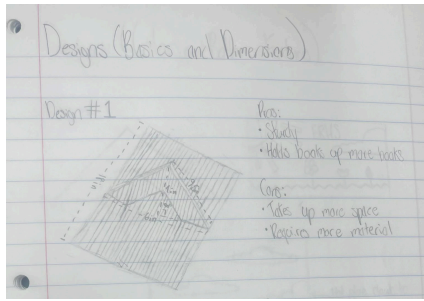
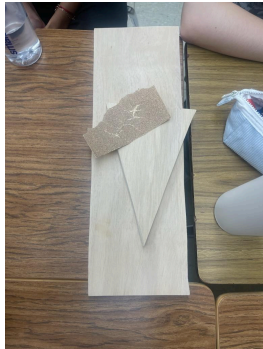


Element G. Construction of a Testable Prototype



Date: Apr 26, 2025

- Initial sketches were made for the design of the bookend and show dimensions.



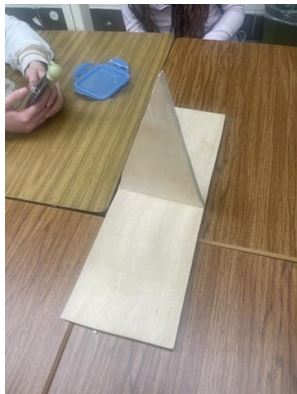
Date: May 1, 2025

- The start of the materials once they were cut and ready to be assembled.



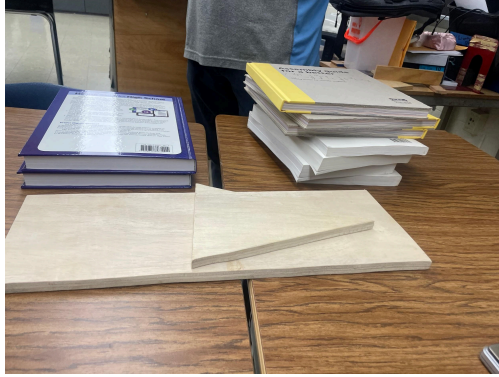
Date: May 1, 2025

- The wood pieces have been assembled into the desired form from the sketches and are being clamped together to allow the glue time to connect the pieces together.



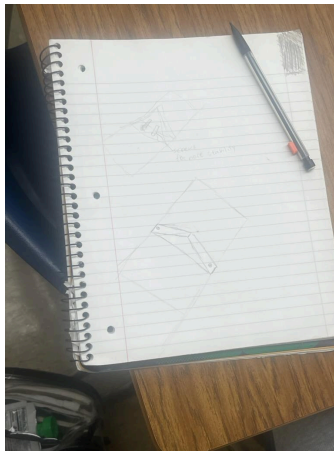
Date: May 2, 2025

- This is the initial prototype ready for testing to see how much force it can take. The wood has been glued together successfully and is holding sturdy.



Date: May 6, 2025

- The initial prototype did not withstand the max test of 5 kg so we had to redesign this picture to show the books that brought it down and how it snapped into its original parts.



Day May 6, 2025

- The sketches of the improved design show the added screws that were added to the bottom of the bookend to keep it from breaking so that it can withstand more force.



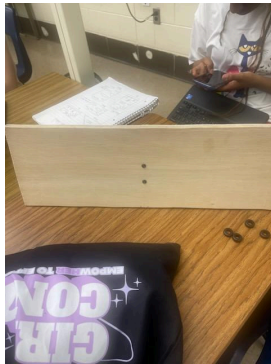
Date: May 7, 2025

- We had just reattached and glued the broken parts together and are preparing to add screws to strengthen the connection between the wood pieces.



Date: May 8, 2025

- This is the top down view of the final prototype where we rebuilt it after the pieces separated from each other due to the initial prototype not being able to withstand the final test of force.



Date May 8, 2025

- This is the bottom view of the final prototype to show how we added screws on the bottom to connect and reinforce our design.

Materials	Description	Units
36"x 24" piece of wood	A long and dense piece of carpenter wood used to make the two wood components of our design	1
screws	Two screws to reinforce our design on the bottom by connecting the two pieces.	2
rubber/ plastic circle parts	4 plastic/rubber mini circular parts that are placed on each corner of the bookend to create friction to prevent movement of the book shelf.	4